

Treat the following exercises:

Exercise 1: (4 pts)

Given the following XML document Edu.xml:

1. <?xml version="1.0" standalone="yes" ?>
2. <Edu Cycle country="LB">
3. <Number n="3">
4. <KG>
5. <Language> FR & EN </Language>
6. <KG1 prerequisite = no > 1 year </KG1>
7. <KG2 prerequisite = "yes" > 1 year's EDU </KG2>
8. <KG3 prerequisite = "yes" > 1 year's EDU </KG3>
9. </KG>
10. <The-Basic-edu>
11. <The_Basic Num="9" prerequisite = "yes" >
12. <Languages>
13. <Ar> required </Ar>
14. <Fr> if Fr is "required" then En is "optional" </Fr>
15. <En> The "Vice versa" </En>
16. </The-Basic-Edu>
17. </The_Basic>
18. <Last:Update>
19. <year> 2014</year>
20. </Last:Update>
21. <notes> the old for the KG1 > or = 3 years</notes>
22. </Edu Cycle>

The XML document Edu.xml is not well formed, for each error:

- Indicate its line number.
- Explain why.
- Propose a correction.

Exercise 2:

Given the following XML document Furniture.xml

```
<?xml version="1.0" standalone="yes" encoding="UTF-8" ?>
<Furniture>
  <Electrical>
    <TV Power="220">
      <Brand> LEGD </Brand>
      <Size> 19 Inches </Size>
      <Model year="2017"> LED-All-HD <Model/>
    </TV>
    <Refrigerator Power="220">
      <Brand> friendship </Brand>
      <Size> 120 feet </Size>
      <Model year="2015"> anti-freeze <Model/>
    </Refrigerator>
```

```

<Micro-wave Power="220">
  <Brand> Cooker </Brand>
  <Size> 10 liter </Size>
  <Model year="2016"> 600 OW <Model/>
</Micro-wave>
</Electrical>
<Non-Electrical>
  <Living-Room>
    <Brand> Picasso </Brand>
    <Size> 1 three sets, 4 one set </Size>
    <Model year="2014"> Italian Style <Model/>
  </Living-Room>
  <Dining-Room>
    <Brand> Good Dining </Brand>
    <Size> Table and 6 chair </Size>
    <Model year="2015"> Lumber Style <Model/>
  </Living-Room>
  <Bed-Room>
    <Brand> Good Night </Brand>
    <Size> Dual bed and 6 parts </Size>
    <Model year="2000"> archetypal Style <Model/>
  </Bed-Room>
</Non-Electrical>
</Furniture>

```

Question 1: (5 pts)

- Write the corresponding DTD file for the Furniture.xml document. (We must use ENTITY)
- In the XML document add the necessary tag, which insure the link to your DTD file.

Question 2: (6 pts)

- Write the corresponding schema file for the Furniture.xml document. (We must use the declaration of a new type).
- In the XML document add the necessary attributes, which insure the link to your schema file.

Question 3: (5 pts)

- Write an XSL file to transform the content of the Furniture.xml document into HTML table (Like the following table in a web site) :

Furniture			
Electrical	Brand	Model	year
TV	LEGD	LED-All-HD	2017
Refrigerator	friendship	anti-freeze	2015
Micro-wave	Cooker	600 OW	2016